



EETS 7304

Module 4 – Part 2 Network Address Translation (NAT) & TCP Detailed Analysis

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SMU®

Topics/Objectives

- Network Address Translation
- Detailed look at TCP Analysis



Network Address Translation (NAT)

Network Address Translation (NAT), Port Address Translation (PAT) and order of operations.



Network Address Translation (NAT)

- NAT is used to conceal the true private IPv4 addresses of a device by using substitute IPv4 addresses in packet headers.
- NAT is usually performed by customer-edge routers, and is more sophisticated today than the older methods of simply using private RFC 1918 addresses whenever one liked.



Typical NAT uses

- You need to connect to the Internet and your hosts do not have globally unique IP addresses.
- You change over to a new ISP that requires you to renumber your network.
- Two intranets with duplicate addresses merge.
- Overlapping networks.



NAT and IPv6

- Once the huge address space in IPv6 is available, there is no need for NAT.



Advantages and Disadvantages of NAT

- Today, NAT still offers advantages, but these often have to be balanced against some disadvantages, especially when coupled with current security practices.



Advantages of NAT

- The advantages to using NAT follow:
 - *Address sharing*—A small number of IP addresses can support a larger pool of devices.
 - *Ease of expansion*—If the number of hosts grows beyond the public IPv4 space assigned, it's easy to add hosts.
 - *Local control*—Administrators essentially run their own private piece of the public Internet.
 - *Easy ISP changeover*—When host addresses are private, public ISP addresses can be changed more easily.
 - *Mainly transparent*—Usually, only a handful of devices have to know the NAT rules for a site.
 - *Security*—Oversold, but still seen as an advantage.
 - Hackers don't know the “real” client's IP address, true, but the true targets are often servers and the NAT “firewalls” themselves.



Disadvantages of NAT

- These NAT pluses have to be balanced against the current list of disadvantages.
 - *Complexity*—NAT adds management complexity and makes even routine troubleshooting more difficult.
 - *Public address sensitivity*—Private addresses are favored by hackers. Some applications and devices raise flags when presented with private addresses.
 - *Application compatibility issues*—NAT is not totally transparent.
 - *Applications* such as FTP, which embed IP addresses and port numbers in data must be handled with special care by NAT Routers.
 - *Poor host accessibility*—NAT makes it difficult to contact local devices from the outside world.
 - NAT is not a good solution for Web sites, FTP servers, or even peer protocols (VoIP) running on a local LAN.
 - *Performance concerns*—The burden of hundreds of simultaneous Internet access users today often degrades NAT router performance for its main task: routing packets.
 - *Security*—Modern protocols such as IPSec raise alarms when packet fields are changed between end systems.
 - You can still combine NAT and IPsec (carefully), but keeping NAT as a “security feature” in addition to IPsec can be tricky.

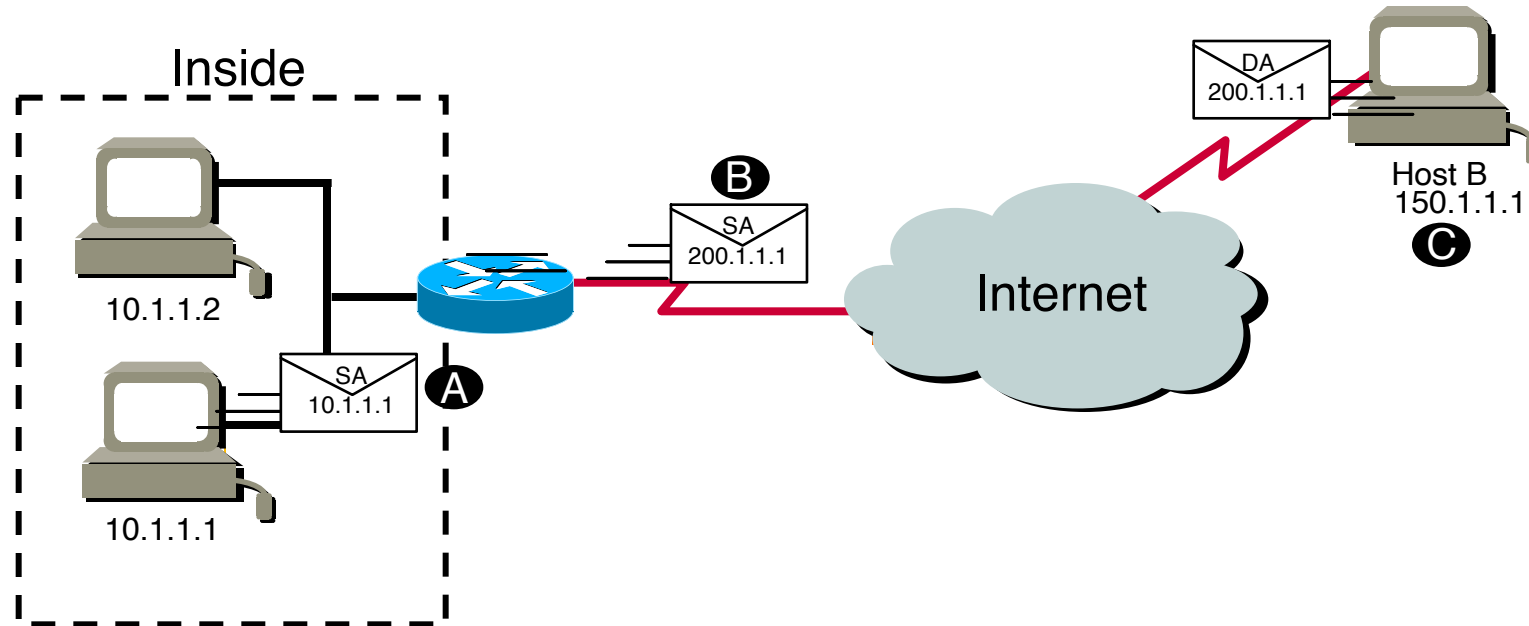


NAT Terms

- **Inside Local Addresses** – An IP address assigned to a host inside a network. This address is likely to be a RFC 1918 private address.
- **Inside Global Address** – A legitimate IP address that represents one or more inside local IP address to the outside world.
- **Outside Global Address** - The IP address assigned to a host on the outside network. The owner of the host assigns this address.
- **Outside Local Address** - The IP address of an outside host as it known to the hosts in the inside network. Usually identical to the outside global address of that outside device.



How NAT Works



NAT table

Inside Global IP Address	Inside Local IP Address	Outside Local IP Address	Outside Global IP Address
200.1.1.1	10.1.1.1	150.1.1.1	150.1.1.1
B	A		C



NAT Functions

- Dynamic NAT
- Dynamic NAT with overloading
- Static NAT
- Translation outside global addresses



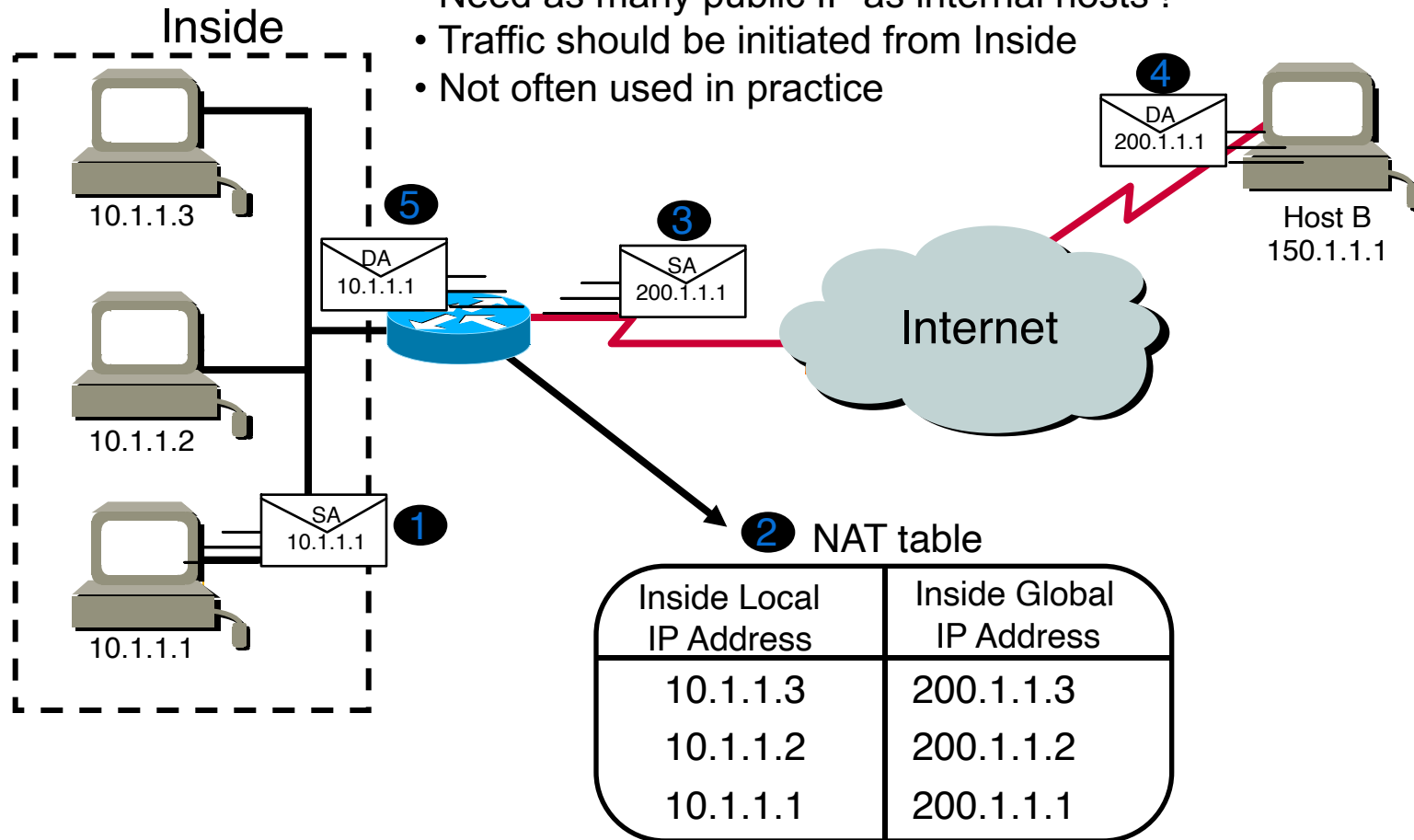
NAT Functions

- Dynamic NAT
 - Uses a pool of public addresses and assigns them on a first-come, first-served basis.
 - When a host with a private IP address requests access to the Internet, dynamic NAT chooses an IP address from the pool that is not already in use by another host and assign it to the host.
- Static NAT
 - Uses one-for-one mapping of local and global addresses and they stay constant. Used for web servers or hosts that must have a consistent IP address that is accessible from the Internet.

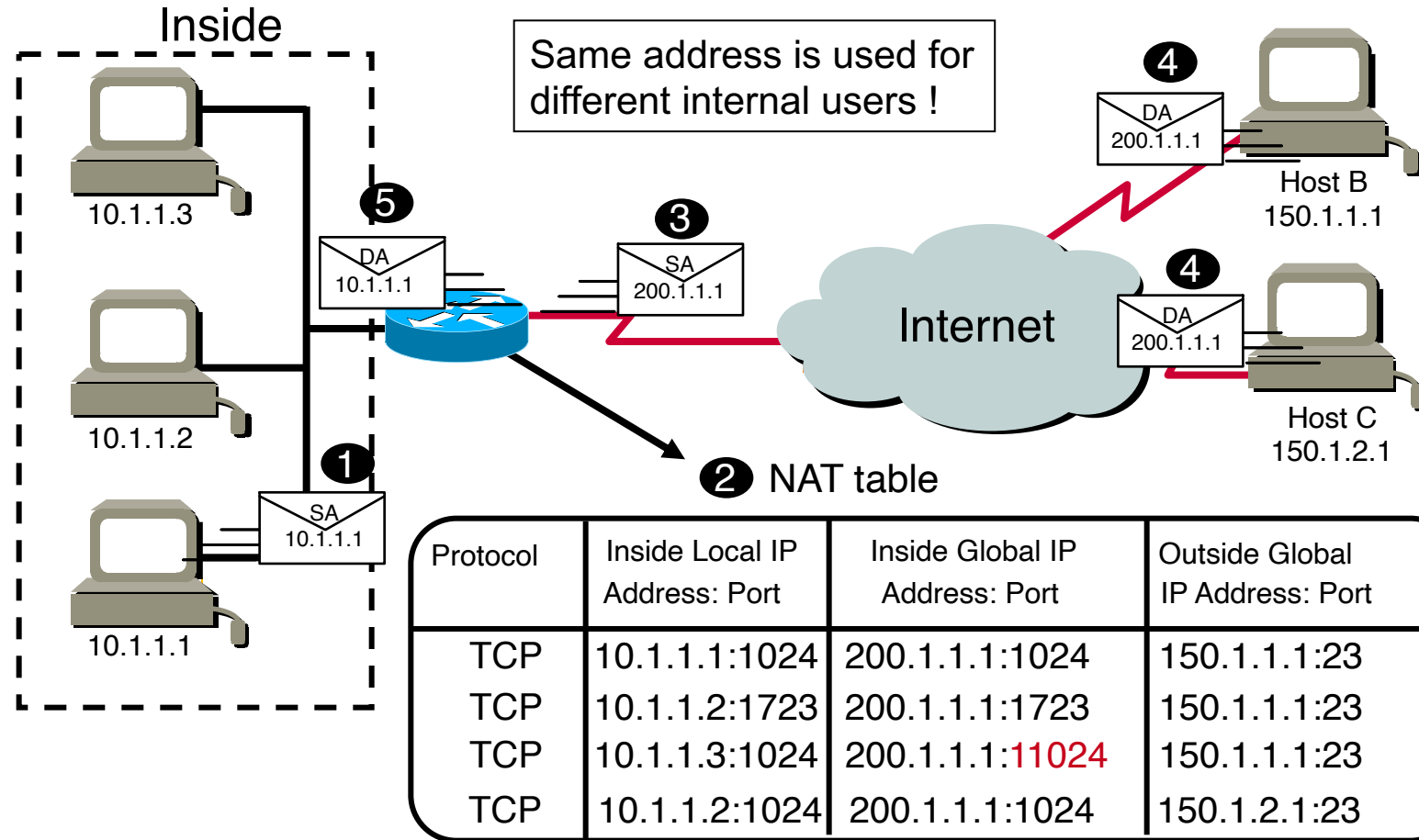


Dynamic NAT: Translating Inside Local Addresses

- A pool of public IP is defined [200.1.1.x]
- Need as many public IP as internal hosts !
- Traffic should be initiated from Inside
- Not often used in practice

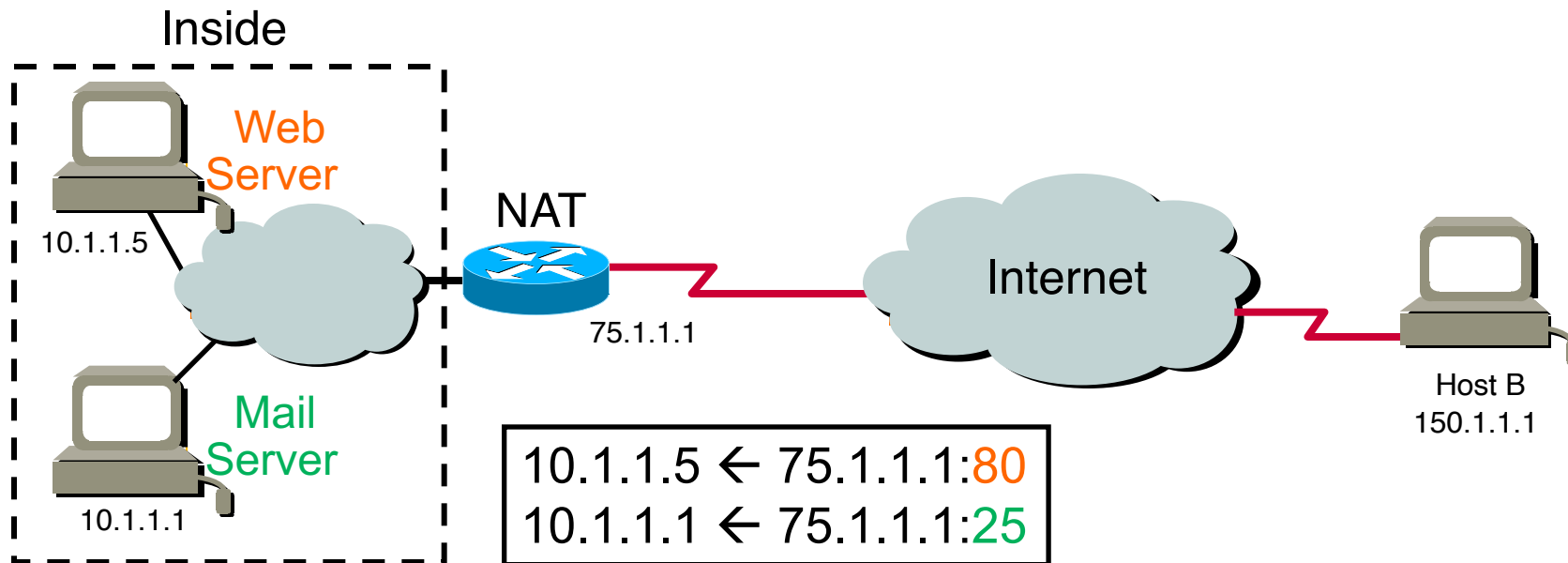


Dynamic NAT with Overloading also called Port Address Translation (PAT)



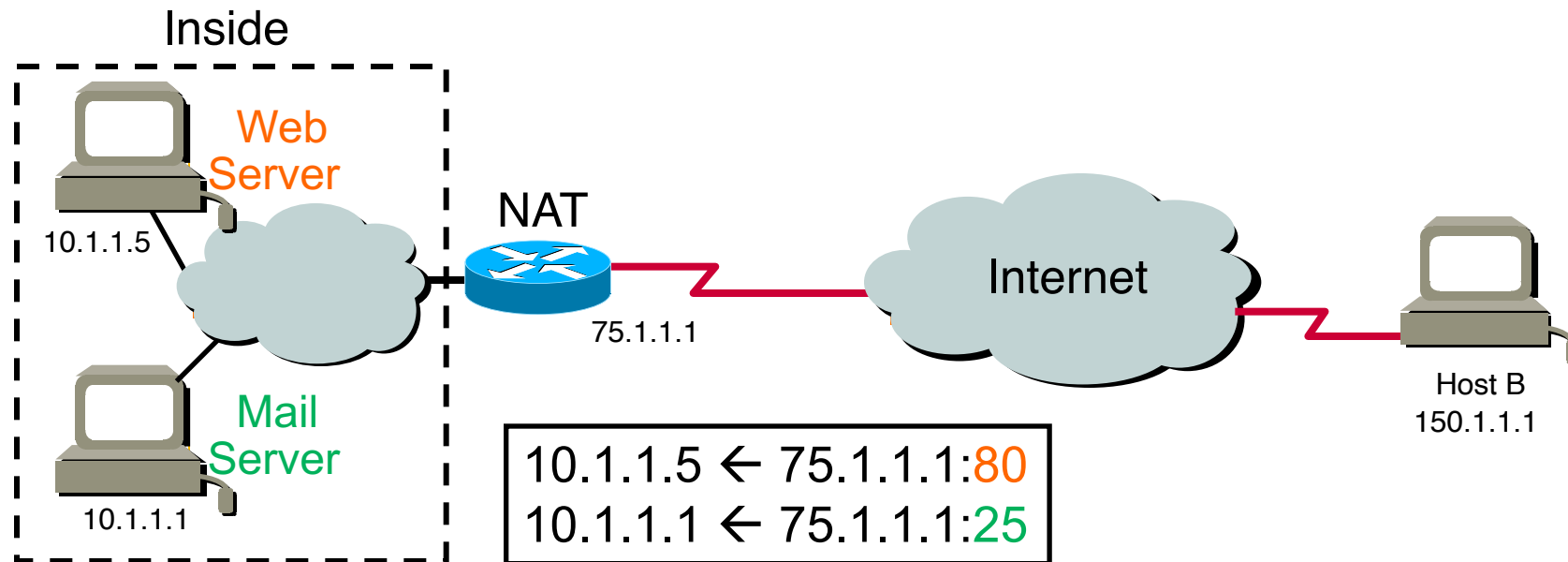
Static NAT: Translating Inside Local Addresses

- Typically used to provide access from Outside to internal servers
- Can map TCP/UDP ports to different Internal servers



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- Typically used to provide access from Outside to internal servers
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Detailed TCP Analysis



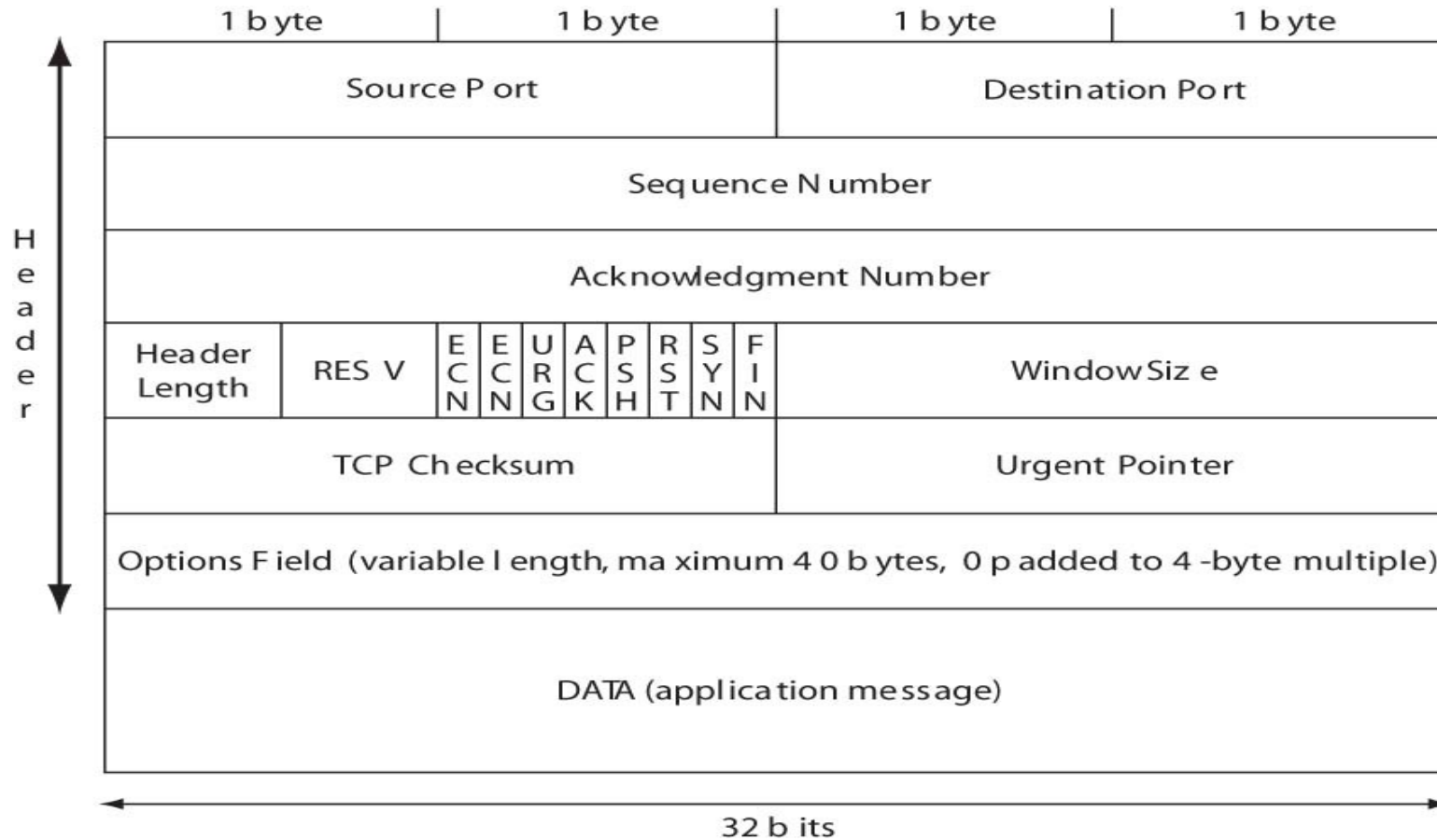
Transmission Control Protocol (TCP) – Review from last week

- The Transmission Control Protocol (TCP) is as complex as UDP is simple.
- Some of the same concepts apply to both because both TCP and UDP are end-to-end protocols.
- Sockets and ports, well-known, dynamic, and private, apply to both.
- The major difference between UDP and TCP is that TCP is connection oriented.

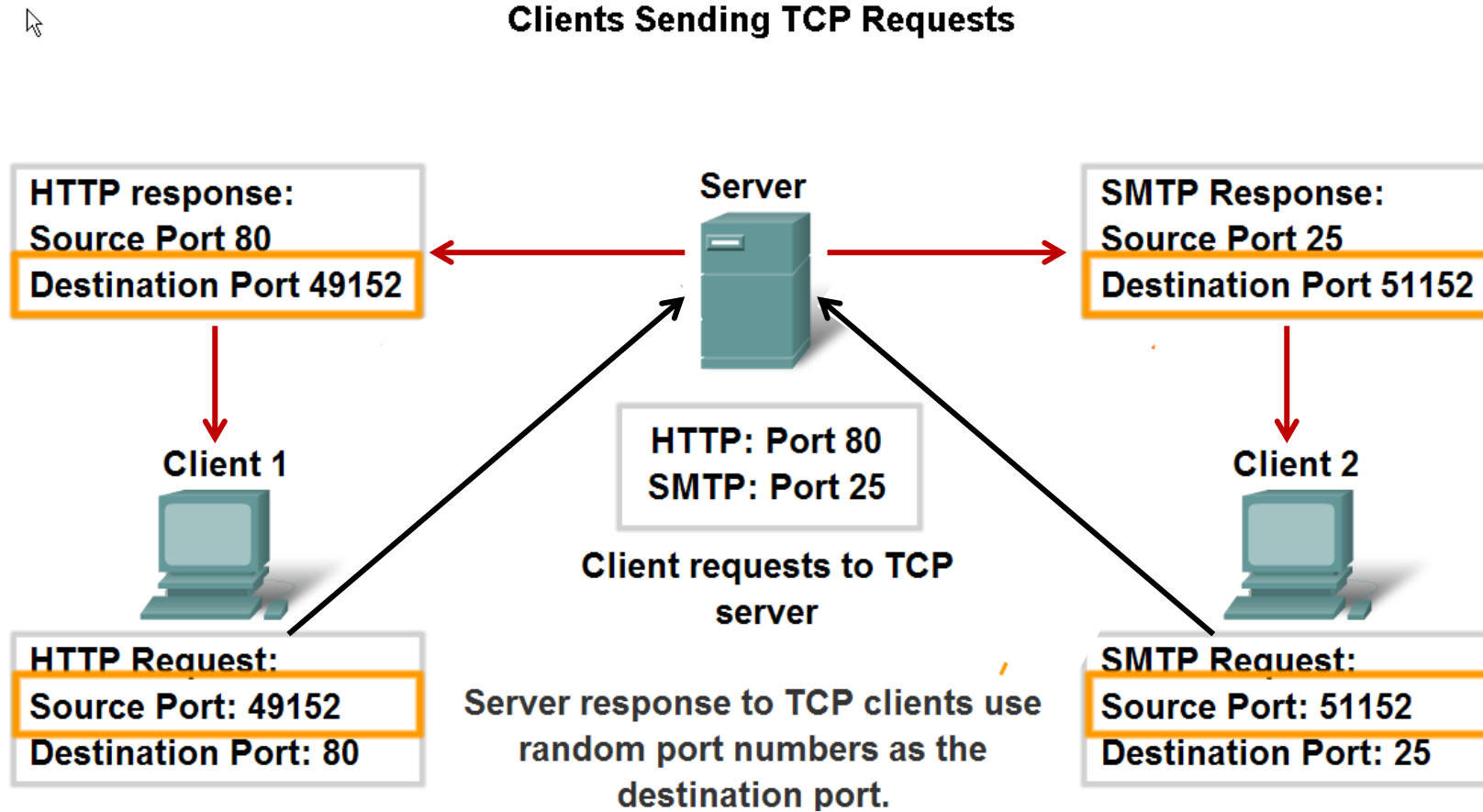


Transmission Control Protocol (TCP) Header

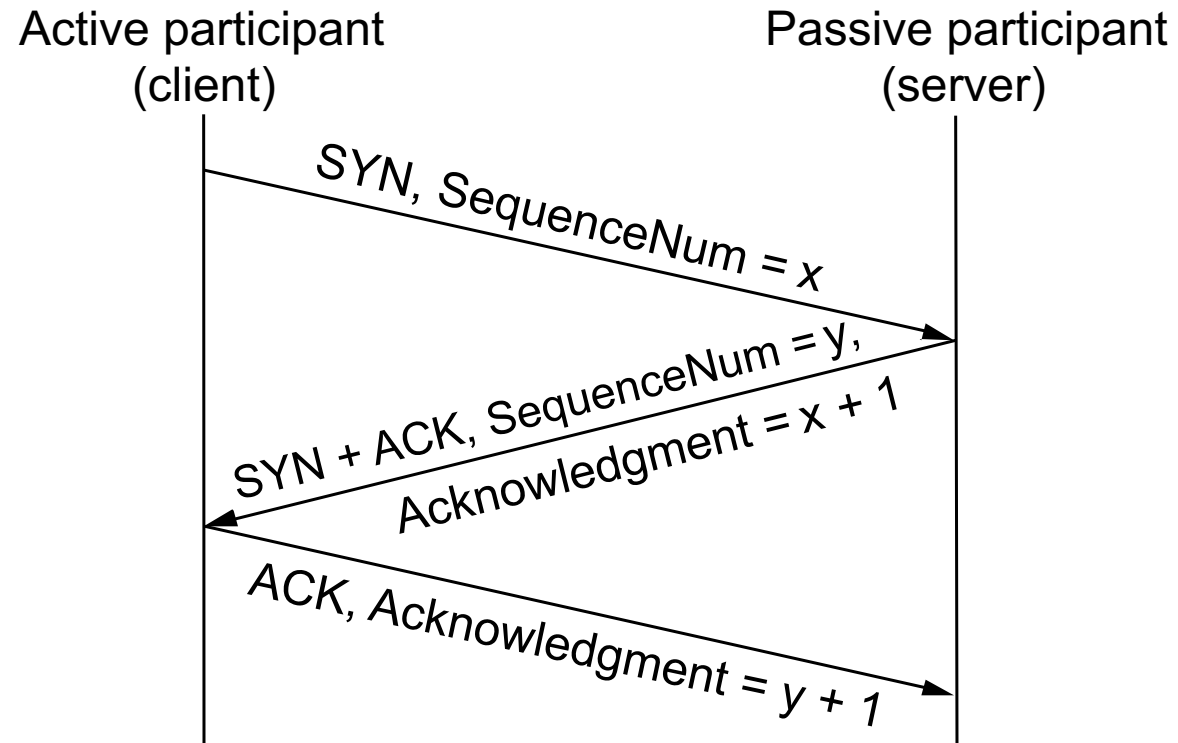
- The TCP header is the same for IPv4 and IPv6



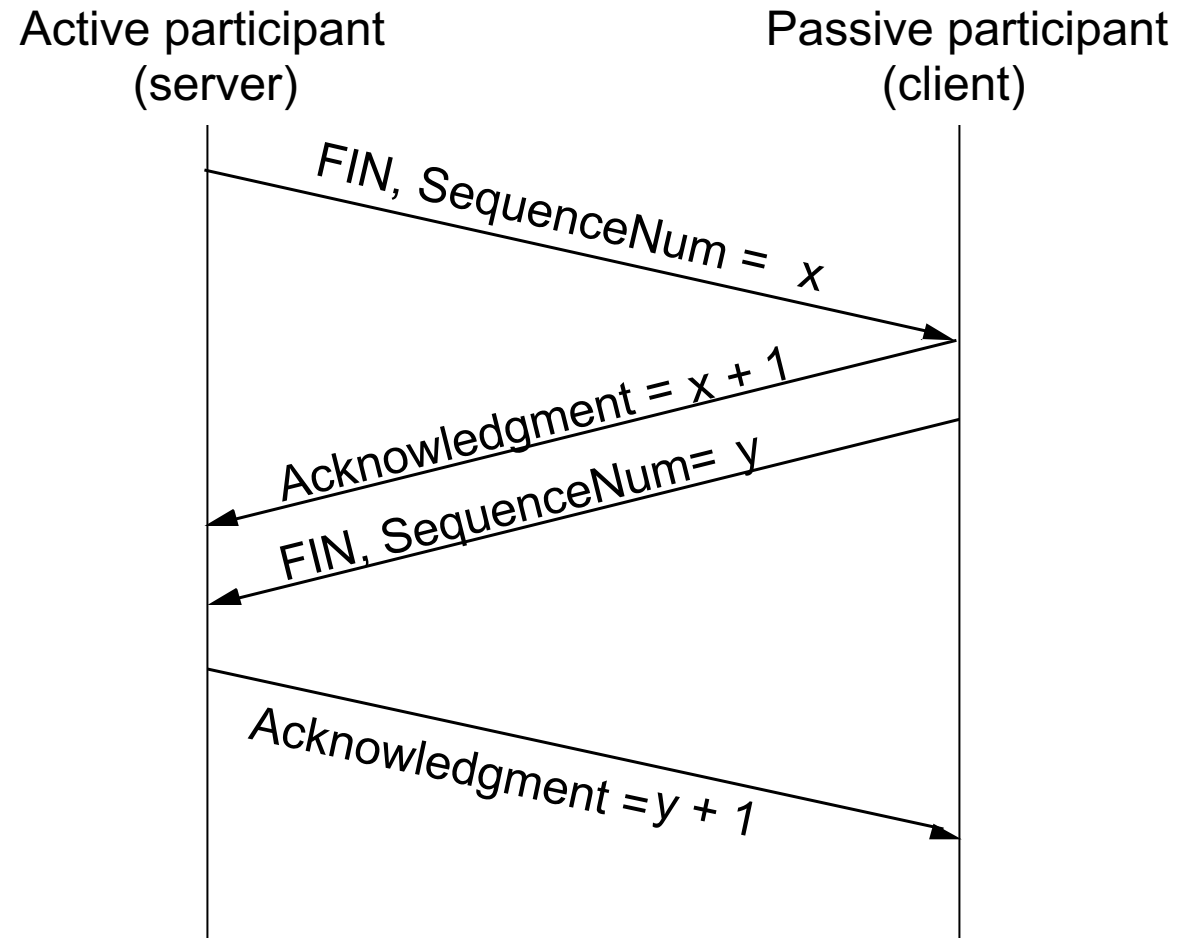
TCP Operation: Ports



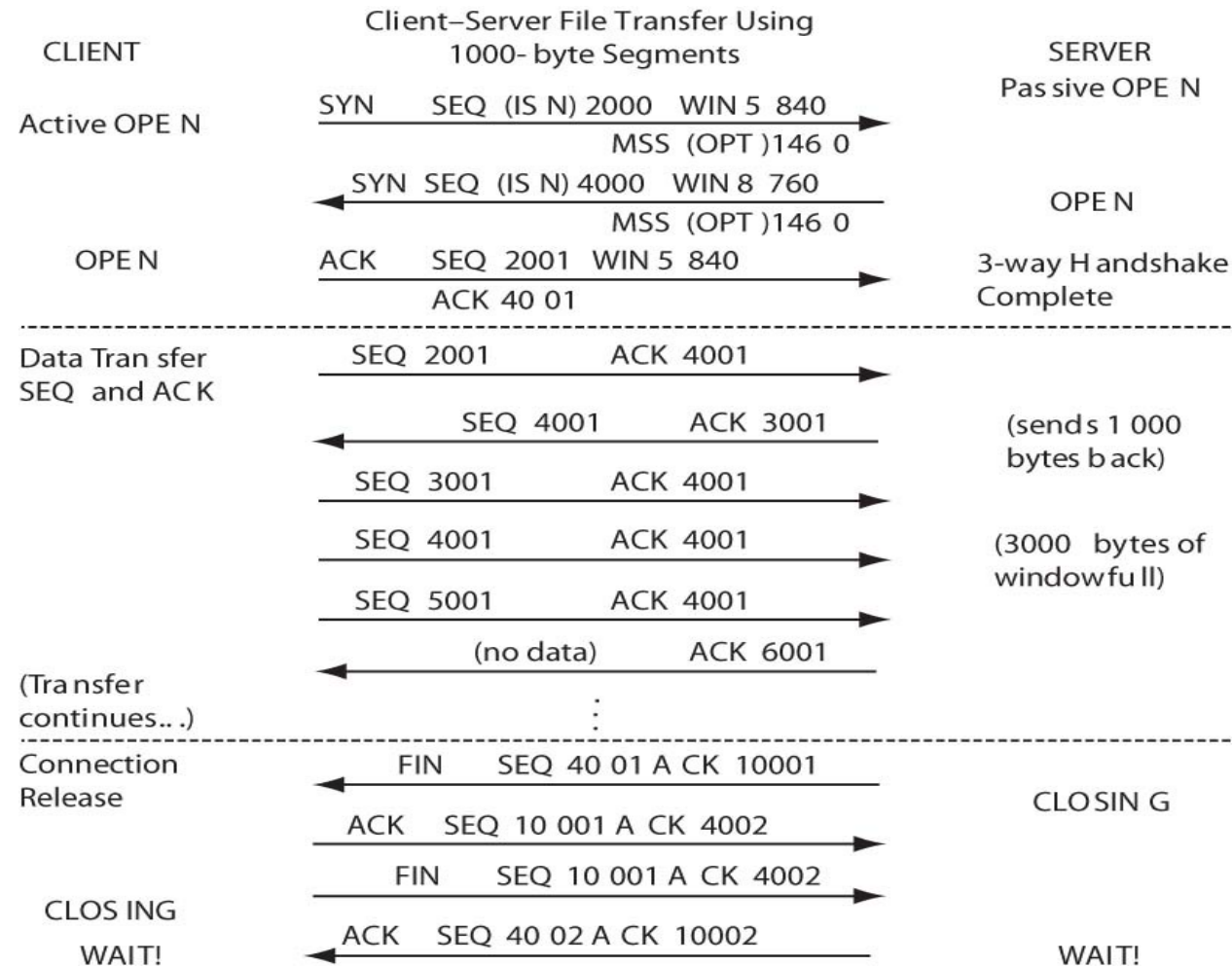
Connection Establishment



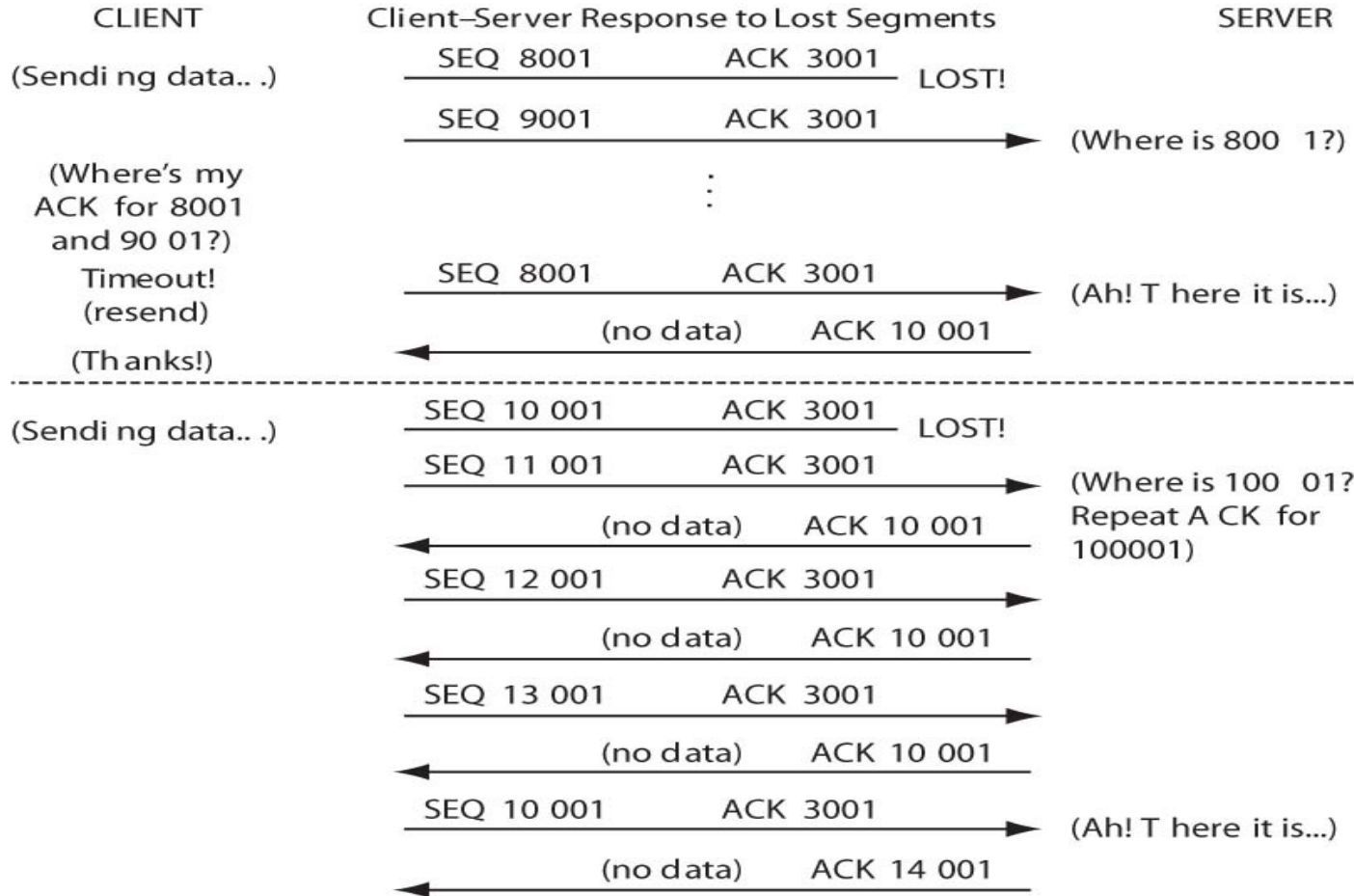
Connection Termination



TCP Operation: Data Exchange



TCP Operation: Data Operation



Summary/Takeaways

- Recognize the types of NAT deployed on the network.
- Characterize how NAT integrates with the order of operations of network routers.
- Identify when NAT/PAT uses both network and transport layers.
- Capture, diagram and analyze a TCP traffic flow that shows the 3-way handshake, and its use of sequence number and acknowledgements.
- Complete Transport Control Protocol (TCP) Lab#7.

